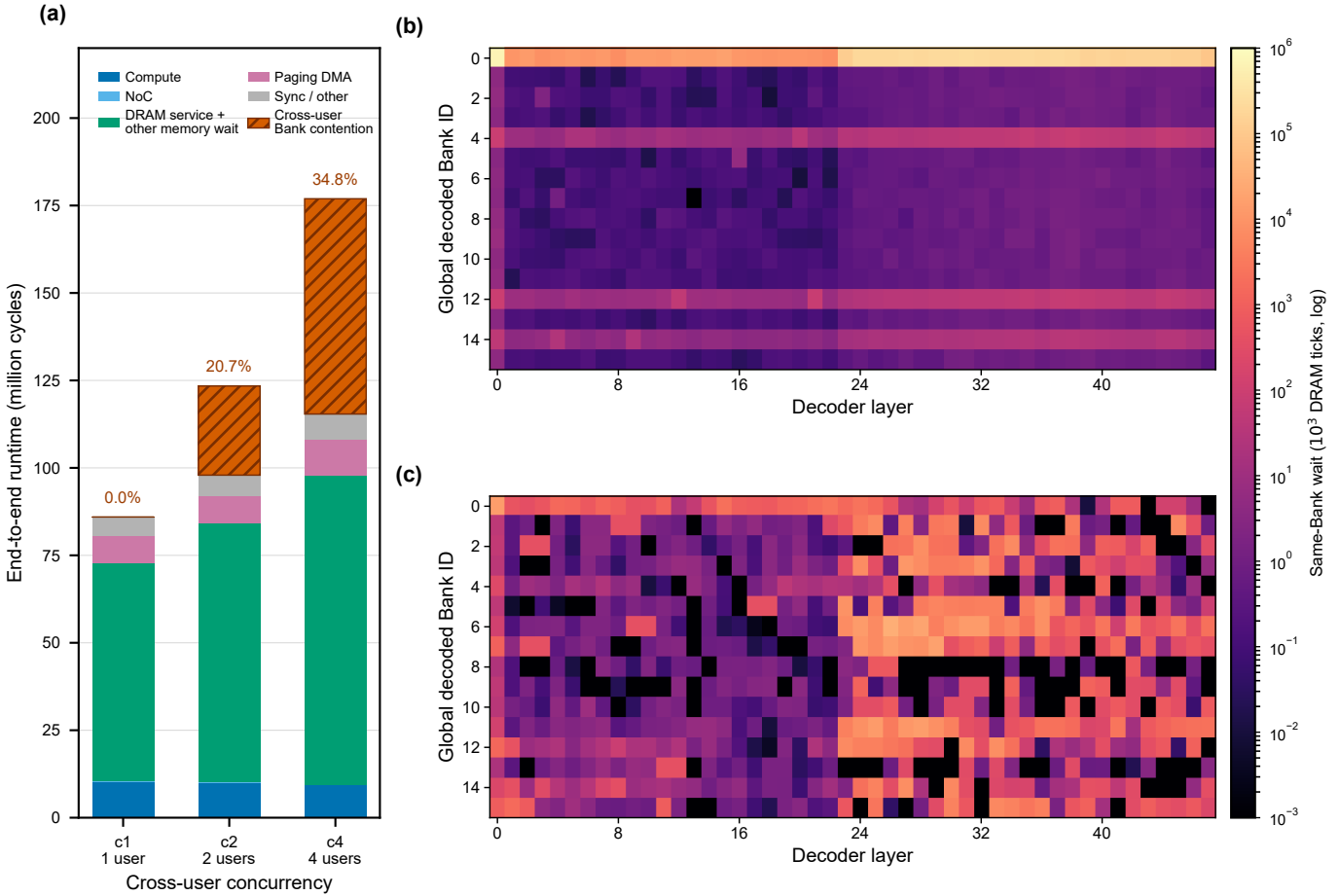

(a)
Compute
NoC
DRAM service + other memory wait
Paging DMA
Sync / other
Cross-user Bank contention
End-to-end runtime (million cycles)
0
25
50
75
100
125
150
175
200
0.0%
20.7%
34.8%
c1
1 user
c2
2 users
c4
4 users
Cross-user concurrency
(b)
Global decoded Bank ID
0
2
4
6
8
10
12
14
0
8
16
24
32
40
Decoder layer
(c)
Global decoded Bank ID
0
2
4
6
8
10
12
14
0
8
16
24
32
40
Decoder layer
Same-Bank wait (10^3 DRAM ticks, log)
10^{6}
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}